Model Architecture: Llama-3.1-8B-Instruct

Generated by sg_modelload

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1 Model Overview

Model name	Llama-3.1-8B-Instruct
Transformers class	`LlamaForCausalLM`
Path	`/models/Llama-3.1-8B-Instruct`
Total parameters	8,030,261,248 (8.03B)
Weight dtype	bfloat16
Hidden size d	4,096
Decoder layers L	32
Query heads H	32
Key/value heads H_{kv}	8
Head dim d_h	128
MLP width d_{ff}	14,336
Vocabulary V	128,256
Max context length	131,072
RMSNorm ϵ	1e-05
Tied input/output embeddings	False
Activation	SiLU
RoPE base θ	500,000

How to read this document. Section 2 writes the whole network as one composed function. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Notation

B, T	batch size, sequence length (tokens)
d	hidden size (4096)
V	vocabulary size (128,256)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 The Model as a Function

Input: token ids $x \in \{0, \ldots, V-1\}^{B \times T}$. Output: next-token logits $y \in \mathbb{R}^{B \times T \times V}$. The 32 decoder layers are written with composition powers, so a block that repeats n times appears once as $\mathcal{B}^{\circ n}$.

$$
\begin{aligned}
h_0 &= \text{Embed}(x) \\
h_L &= \left(\mathcal{B}_{\text{D}}^{\circ 32}\right)(h_0) \\
y &= W_{\text{LM}}\ \text{RMSNorm}_{\text{final}}(h_L)
\end{aligned}
$$

2.1 Layer pattern

Symbol	Block type	Count	Params / block	Layer indices
$\mathcal{B}_{\text{D}}$	Decoder block	32	218,112,000	0–31

Language-model decoder (32 layers). layers 0–31: $[\mathcal{B}_{\text{D}}]$ repeated 32×.

2.2 Block functions

$\mathcal{B}_{\mathrm{D}}$ — **Decoder block.**

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
Q &= W_Q\, u && H = 32 \text{ heads of size } d_h = 128\\
K,\ V &= W_K\, u, \quad W_V\, u && H_{kv} = 8 \text{ key/value heads}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\,[O_1; \ldots; O_H]\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\mathrm{SiLU}(W_{\mathrm{gate}}\, u) \odot W_{\mathrm{up}}\, u\big) && \text{gated MLP, width 14336}
\end{aligned}
$$

3 Forward Pass in Pseudocode

Algorithm 1 Llama-3.1-8B-Instruct: full forward pass

Require: token ids x of shape $B \times T$

$h \leftarrow \mathrm{Embed}(x)$ ▷ look up one vector per token: $B{\times}T{\times}4096$

for $r \leftarrow 0$ **to** 31 **do** ▷ layers 0–31: pattern of 1 repeated 32 times

  $h \leftarrow \mathcal{B}_{\mathrm{D}}(h)$ ▷ Decoder block

end for

$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$

$y \leftarrow h\, W_{\mathrm{LM}}^{\top}$ ▷ logits $B{\times}T{\times}128,256$

return y ▷ softmax(y) = next-token probabilities

Algorithm 2 Block $\mathcal{B}_{\mathrm{D}}(h)$ — Decoder block

Require: h of shape $B \times T \times 4096$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection

$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector

token mixing: multi-head self-attention

$Q \leftarrow uW_Q^{\top}$ ▷ $B{\times}T{\times}32{\times}128$ (32 query heads)

$K \leftarrow uW_K^{\top};\ V \leftarrow uW_V^{\top}$ ▷ $B{\times}T{\times}8{\times}128$ (8 key/value heads, each shared by 4 query heads)

$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs

$P \leftarrow \mathrm{softmax}(s\, QK^{\top} + M)$ ▷ mask: causal: a token sees only earlier tokens

$O \leftarrow PV$ ▷ $B{\times}T{\times}32{\times}128$

$m \leftarrow O\, W_O^{\top}$ ▷ concatenate heads, project back to $B{\times}T{\times}4096$

$h \leftarrow r + m$ ▷ add residual

channel mixing: feed-forward network (each token independently)

$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$

$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^{\top}) \odot uW_{\mathrm{up}}^{\top}\big)W_{\mathrm{down}}^{\top}$ ▷ expand to 14336, gate, shrink back to 4096

$h \leftarrow r + m$ ▷ add residual

return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 2: Weights that appear once

Weight	Type	Shape	Params
`model.embed_tokens.weight`	Embedding	128256×4096	525,336,576
`model.norm.weight`	LlamaRMSNorm	4096	4,096

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Weight	Type	Shape	Params
`lm_head.weight`	Linear	128256×4096	525,336,576

4.2 Language-model decoder: `model.layers`

Table 3: $\mathcal{B}_{\mathrm{D}}$: Decoder block — layers 0–31; weight names are relative to `model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	4096×4096	16,777,216
`self_attn.k_proj.weight`	Linear	1024×4096	4,194,304
`self_attn.v_proj.weight`	Linear	1024×4096	4,194,304
`self_attn.o_proj.weight`	Linear	4096×4096	16,777,216
`mlp.gate_proj.weight`	Linear	14336×4096	58,720,256
`mlp.up_proj.weight`	Linear	14336×4096	58,720,256
`mlp.down_proj.weight`	Linear	4096×14336	58,720,256
`input_layernorm.weight`	LlamaRMSNorm	4096	4,096
`post_attention_layernorm.weight`	LlamaRMSNorm	4096	4,096
Per block			**218,112,000**
× 32 blocks			**6,979,584,000**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
Language model	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
LM head (output projection) `lm_head`	Maps each final hidden vector to one score (logit) per vocabulary token. Softmax of the logits gives the next-token probabilities. When tied, it reuses the embedding matrix.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.

6 Parameter Count

Table 5: Parameters by component

Component	Parameters	% of total
MLP / MoE	5,637,144,576	70.2%
Attention / token mixer	1,342,177,280	16.7%
Token embedding	525,336,576	6.5%
LM head	525,336,576	6.5%
Norms	266,240	0.0%
Total	**8,030,261,248**	**100.0%**

Table 6: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{D}}$ Decoder block	32	218,112,000	6,979,584,000
Modules outside stacks	1	–	1,050,677,248
Total			**8,030,261,248**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight`	Linear	128256 × 4096	bfloat16	525,336,576
`model`	LlamaModel	–	–	–
`model.embed_tokens.weight`	Embedding	128256 × 4096	bfloat16	525,336,576
`model.layers.0`	LlamaDecoderLayer	–	–	–
`model.layers.0.input_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.0.mlp`	LlamaMLP	–	–	–
`model.layers.0.mlp.down_proj.weight`	Linear	4096 × 14336	bfloat16	58,720,256
`model.layers.0.mlp.gate_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.0.mlp.up_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.0.post_attention_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.0.self_attn`	LlamaAttention	–	–	–
`model.layers.0.self_attn.k_proj.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.layers.0.self_attn.o_proj.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.layers.0.self_attn.q_proj.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.layers.0.self_attn.v_proj.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.layers.1`	LlamaDecoderLayer	–	–	–
`model.layers.1.input_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.1.mlp`	LlamaMLP	–	–	–
`model.layers.1.mlp.down_proj.weight`	Linear	4096 × 14336	bfloat16	58,720,256
`model.layers.1.mlp.gate_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.1.mlp.up_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.1.post_attention_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.1.self_attn`	LlamaAttention	–	–	–
`model.layers.1.self_attn.k_proj.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.layers.1.self_attn.o_proj.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.layers.1.self_attn.q_proj.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.layers.1.self_attn.v_proj.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.layers.2`	LlamaDecoderLayer	–	–	–
`model.layers.2.input_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.2.mlp`	LlamaMLP	–	–	–
`model.layers.2.mlp.down_proj.weight`	Linear	4096 × 14336	bfloat16	58,720,256
`model.layers.2.mlp.gate_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.2.mlp.up_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.2.post_attention_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.2.self_attn`	LlamaAttention	–	–	–
`model.layers.2.self_attn.k_proj.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.layers.2.self_attn.o_proj.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.layers.2.self_attn.q_proj.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.layers.2.self_attn.v_proj.weight`	Linear	1024 × 4096	bfloat16	4,194,304

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Layer path	Type	Shape	Dtype	Params
model.layers.3	LlamaDecoderLayer	–	–	–
model.layers.3.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.3.mlp	LlamaMLP	–	–	–
model.layers.3.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.3.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.3.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.3.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.3.self_attn	LlamaAttention	–	–	–
model.layers.3.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.3.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.3.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.3.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.4	LlamaDecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.4.mlp	LlamaMLP	–	–	–
model.layers.4.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.4.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.4.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.4.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.4.self_attn	LlamaAttention	–	–	–
model.layers.4.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.4.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.4.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.4.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.5	LlamaDecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.5.mlp	LlamaMLP	–	–	–
model.layers.5.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.5.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.5.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.5.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.5.self_attn	LlamaAttention	–	–	–
model.layers.5.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.5.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.5.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.5.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.6	LlamaDecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.6.mlp	LlamaMLP	–	–	–
model.layers.6.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.6.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.6.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.6.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.6.self_attn	LlamaAttention	–	–	–
model.layers.6.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.6.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.6.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.6.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.7	LlamaDecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.7.mlp	LlamaMLP	–	–	–
model.layers.7.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.7.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.7.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.7.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.7.self_attn	LlamaAttention	–	–	–
model.layers.7.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.7.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.7.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.7.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.8	LlamaDecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.8.mlp	LlamaMLP	–	–	–
model.layers.8.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.8.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.8.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.8.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.8.self_attn	LlamaAttention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.layers.8.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.8.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.8.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.8.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.9	LlamaDecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.9.mlp	LlamaMLP	–	–	–
model.layers.9.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.9.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.9.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.9.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.9.self_attn	LlamaAttention	–	–	–
model.layers.9.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.9.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.9.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.9.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.10	LlamaDecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.10.mlp	LlamaMLP	–	–	–
model.layers.10.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.10.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.10.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.10.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.10.self_attn	LlamaAttention	–	–	–
model.layers.10.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.10.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.10.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.10.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.11	LlamaDecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.11.mlp	LlamaMLP	–	–	–
model.layers.11.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.11.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.11.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.11.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.11.self_attn	LlamaAttention	–	–	–
model.layers.11.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.11.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.11.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.11.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.12	LlamaDecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.12.mlp	LlamaMLP	–	–	–
model.layers.12.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.12.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.12.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.12.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.12.self_attn	LlamaAttention	–	–	–
model.layers.12.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.12.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.12.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.12.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.13	LlamaDecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.13.mlp	LlamaMLP	–	–	–
model.layers.13.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.13.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.13.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.13.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.13.self_attn	LlamaAttention	–	–	–
model.layers.13.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.13.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.13.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.13.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.14	LlamaDecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.14.mlp	LlamaMLP	–	–	–
model.layers.14.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.14.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256

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Layer path	Type	Shape	Dtype	Params
model.layers.14.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.14.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.14.self_attn	LlamaAttention	–	–	–
model.layers.14.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.14.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.14.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.14.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.15	LlamaDecoderLayer	–	–	–
model.layers.15.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.15.mlp	LlamaMLP	–	–	–
model.layers.15.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.15.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.15.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.15.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.15.self_attn	LlamaAttention	–	–	–
model.layers.15.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.15.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.15.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.15.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.16	LlamaDecoderLayer	–	–	–
model.layers.16.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.16.mlp	LlamaMLP	–	–	–
model.layers.16.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.16.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.16.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.16.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.16.self_attn	LlamaAttention	–	–	–
model.layers.16.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.16.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.16.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.16.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.17	LlamaDecoderLayer	–	–	–
model.layers.17.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.17.mlp	LlamaMLP	–	–	–
model.layers.17.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.17.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.17.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.17.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.17.self_attn	LlamaAttention	–	–	–
model.layers.17.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.17.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.17.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.17.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.18	LlamaDecoderLayer	–	–	–
model.layers.18.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.18.mlp	LlamaMLP	–	–	–
model.layers.18.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.18.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.18.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.18.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.18.self_attn	LlamaAttention	–	–	–
model.layers.18.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.18.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.18.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.18.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.19	LlamaDecoderLayer	–	–	–
model.layers.19.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.19.mlp	LlamaMLP	–	–	–
model.layers.19.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.19.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.19.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.19.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.19.self_attn	LlamaAttention	–	–	–
model.layers.19.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.19.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.19.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.19.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.20	LlamaDecoderLayer	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.layers.20.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.20.mlp	LlamaMLP	–	–	–
model.layers.20.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.20.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.20.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.20.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.20.self_attn	LlamaAttention	–	–	–
model.layers.20.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.20.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.20.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.20.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.21	LlamaDecoderLayer	–	–	–
model.layers.21.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.21.mlp	LlamaMLP	–	–	–
model.layers.21.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.21.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.21.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.21.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.21.self_attn	LlamaAttention	–	–	–
model.layers.21.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.21.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.21.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.21.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.22	LlamaDecoderLayer	–	–	–
model.layers.22.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.22.mlp	LlamaMLP	–	–	–
model.layers.22.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.22.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.22.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.22.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.22.self_attn	LlamaAttention	–	–	–
model.layers.22.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.22.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.22.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.22.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.23	LlamaDecoderLayer	–	–	–
model.layers.23.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.23.mlp	LlamaMLP	–	–	–
model.layers.23.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.23.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.23.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.23.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.23.self_attn	LlamaAttention	–	–	–
model.layers.23.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.23.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.23.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.23.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.24	LlamaDecoderLayer	–	–	–
model.layers.24.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.24.mlp	LlamaMLP	–	–	–
model.layers.24.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.24.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.24.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.24.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.24.self_attn	LlamaAttention	–	–	–
model.layers.24.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.24.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.24.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.24.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.25	LlamaDecoderLayer	–	–	–
model.layers.25.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.25.mlp	LlamaMLP	–	–	–
model.layers.25.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.25.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.25.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.25.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.25.self_attn	LlamaAttention	–	–	–
model.layers.25.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304

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Layer path	Type	Shape	Dtype	Params
model.layers.25.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.25.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.25.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.26	LlamaDecoderLayer	–	–	–
model.layers.26.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.26.mlp	LlamaMLP	–	–	–
model.layers.26.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.26.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.26.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.26.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.26.self_attn	LlamaAttention	–	–	–
model.layers.26.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.26.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.26.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.26.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.27	LlamaDecoderLayer	–	–	–
model.layers.27.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.27.mlp	LlamaMLP	–	–	–
model.layers.27.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.27.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.27.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.27.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.27.self_attn	LlamaAttention	–	–	–
model.layers.27.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.27.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.27.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.27.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.28	LlamaDecoderLayer	–	–	–
model.layers.28.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.28.mlp	LlamaMLP	–	–	–
model.layers.28.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.28.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.28.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.28.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.28.self_attn	LlamaAttention	–	–	–
model.layers.28.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.28.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.28.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.28.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.29	LlamaDecoderLayer	–	–	–
model.layers.29.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.29.mlp	LlamaMLP	–	–	–
model.layers.29.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.29.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.29.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.29.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.29.self_attn	LlamaAttention	–	–	–
model.layers.29.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.29.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.29.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.29.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.30	LlamaDecoderLayer	–	–	–
model.layers.30.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.30.mlp	LlamaMLP	–	–	–
model.layers.30.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.30.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.30.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.30.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.30.self_attn	LlamaAttention	–	–	–
model.layers.30.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.30.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.30.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.30.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.31	LlamaDecoderLayer	–	–	–
model.layers.31.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.31.mlp	LlamaMLP	–	–	–
model.layers.31.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.31.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.31.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256

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Layer path	Type	Shape	Dtype	Params
model.layers.31.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.31.self_attn	LlamaAttention	–	–	–
model.layers.31.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.31.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.31.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.31.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.norm.weight	LlamaRMSNorm	4096	bfloat16	4,096

Appendix B Per-Layer Parameter Count

Module	Params	Running total
lm_head	525,336,576	525,336,576
model.embed_tokens	525,336,576	1,050,673,152
model.layers.0.input_layernorm	4,096	1,050,677,248
model.layers.0.mlp.down_proj	58,720,256	1,109,397,504
model.layers.0.mlp.gate_proj	58,720,256	1,168,117,760
model.layers.0.mlp.up_proj	58,720,256	1,226,838,016
model.layers.0.post_attention_layernorm	4,096	1,226,842,112
model.layers.0.self_attn.k_proj	4,194,304	1,231,036,416
model.layers.0.self_attn.o_proj	16,777,216	1,247,813,632
model.layers.0.self_attn.q_proj	16,777,216	1,264,590,848
model.layers.0.self_attn.v_proj	4,194,304	1,268,785,152
model.layers.1.input_layernorm	4,096	1,268,789,248
model.layers.1.mlp.down_proj	58,720,256	1,327,509,504
model.layers.1.mlp.gate_proj	58,720,256	1,386,229,760
model.layers.1.mlp.up_proj	58,720,256	1,444,950,016
model.layers.1.post_attention_layernorm	4,096	1,444,954,112
model.layers.1.self_attn.k_proj	4,194,304	1,449,148,416
model.layers.1.self_attn.o_proj	16,777,216	1,465,925,632
model.layers.1.self_attn.q_proj	16,777,216	1,482,702,848
model.layers.1.self_attn.v_proj	4,194,304	1,486,897,152
model.layers.2.input_layernorm	4,096	1,486,901,248
model.layers.2.mlp.down_proj	58,720,256	1,545,621,504
model.layers.2.mlp.gate_proj	58,720,256	1,604,341,760
model.layers.2.mlp.up_proj	58,720,256	1,663,062,016
model.layers.2.post_attention_layernorm	4,096	1,663,066,112
model.layers.2.self_attn.k_proj	4,194,304	1,667,260,416
model.layers.2.self_attn.o_proj	16,777,216	1,684,037,632
model.layers.2.self_attn.q_proj	16,777,216	1,700,814,848
model.layers.2.self_attn.v_proj	4,194,304	1,705,009,152
model.layers.3.input_layernorm	4,096	1,705,013,248
model.layers.3.mlp.down_proj	58,720,256	1,763,733,504
model.layers.3.mlp.gate_proj	58,720,256	1,822,453,760
model.layers.3.mlp.up_proj	58,720,256	1,881,174,016
model.layers.3.post_attention_layernorm	4,096	1,881,178,112
model.layers.3.self_attn.k_proj	4,194,304	1,885,372,416
model.layers.3.self_attn.o_proj	16,777,216	1,902,149,632
model.layers.3.self_attn.q_proj	16,777,216	1,918,926,848
model.layers.3.self_attn.v_proj	4,194,304	1,923,121,152
model.layers.4.input_layernorm	4,096	1,923,125,248
model.layers.4.mlp.down_proj	58,720,256	1,981,845,504
model.layers.4.mlp.gate_proj	58,720,256	2,040,565,760
model.layers.4.mlp.up_proj	58,720,256	2,099,286,016
model.layers.4.post_attention_layernorm	4,096	2,099,290,112
model.layers.4.self_attn.k_proj	4,194,304	2,103,484,416
model.layers.4.self_attn.o_proj	16,777,216	2,120,261,632
model.layers.4.self_attn.q_proj	16,777,216	2,137,038,848

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Module	Params	Running total
model.layers.4.self_attn.v_proj	4,194,304	2,141,233,152
model.layers.5.input_layernorm	4,096	2,141,237,248
model.layers.5.mlp.down_proj	58,720,256	2,199,957,504
model.layers.5.mlp.gate_proj	58,720,256	2,258,677,760
model.layers.5.mlp.up_proj	58,720,256	2,317,398,016
model.layers.5.post_attention_layernorm	4,096	2,317,402,112
model.layers.5.self_attn.k_proj	4,194,304	2,321,596,416
model.layers.5.self_attn.o_proj	16,777,216	2,338,373,632
model.layers.5.self_attn.q_proj	16,777,216	2,355,150,848
model.layers.5.self_attn.v_proj	4,194,304	2,359,345,152
model.layers.6.input_layernorm	4,096	2,359,349,248
model.layers.6.mlp.down_proj	58,720,256	2,418,069,504
model.layers.6.mlp.gate_proj	58,720,256	2,476,789,760
model.layers.6.mlp.up_proj	58,720,256	2,535,510,016
model.layers.6.post_attention_layernorm	4,096	2,535,514,112
model.layers.6.self_attn.k_proj	4,194,304	2,539,708,416
model.layers.6.self_attn.o_proj	16,777,216	2,556,485,632
model.layers.6.self_attn.q_proj	16,777,216	2,573,262,848
model.layers.6.self_attn.v_proj	4,194,304	2,577,457,152
model.layers.7.input_layernorm	4,096	2,577,461,248
model.layers.7.mlp.down_proj	58,720,256	2,636,181,504
model.layers.7.mlp.gate_proj	58,720,256	2,694,901,760
model.layers.7.mlp.up_proj	58,720,256	2,753,622,016
model.layers.7.post_attention_layernorm	4,096	2,753,626,112
model.layers.7.self_attn.k_proj	4,194,304	2,757,820,416
model.layers.7.self_attn.o_proj	16,777,216	2,774,597,632
model.layers.7.self_attn.q_proj	16,777,216	2,791,374,848
model.layers.7.self_attn.v_proj	4,194,304	2,795,569,152
model.layers.8.input_layernorm	4,096	2,795,573,248
model.layers.8.mlp.down_proj	58,720,256	2,854,293,504
model.layers.8.mlp.gate_proj	58,720,256	2,913,013,760
model.layers.8.mlp.up_proj	58,720,256	2,971,734,016
model.layers.8.post_attention_layernorm	4,096	2,971,738,112
model.layers.8.self_attn.k_proj	4,194,304	2,975,932,416
model.layers.8.self_attn.o_proj	16,777,216	2,992,709,632
model.layers.8.self_attn.q_proj	16,777,216	3,009,486,848
model.layers.8.self_attn.v_proj	4,194,304	3,013,681,152
model.layers.9.input_layernorm	4,096	3,013,685,248
model.layers.9.mlp.down_proj	58,720,256	3,072,405,504
model.layers.9.mlp.gate_proj	58,720,256	3,131,125,760
model.layers.9.mlp.up_proj	58,720,256	3,189,846,016
model.layers.9.post_attention_layernorm	4,096	3,189,850,112
model.layers.9.self_attn.k_proj	4,194,304	3,194,044,416
model.layers.9.self_attn.o_proj	16,777,216	3,210,821,632
model.layers.9.self_attn.q_proj	16,777,216	3,227,598,848
model.layers.9.self_attn.v_proj	4,194,304	3,231,793,152
model.layers.10.input_layernorm	4,096	3,231,797,248
model.layers.10.mlp.down_proj	58,720,256	3,290,517,504
model.layers.10.mlp.gate_proj	58,720,256	3,349,237,760
model.layers.10.mlp.up_proj	58,720,256	3,407,958,016
model.layers.10.post_attention_layernorm	4,096	3,407,962,112
model.layers.10.self_attn.k_proj	4,194,304	3,412,156,416
model.layers.10.self_attn.o_proj	16,777,216	3,428,933,632
model.layers.10.self_attn.q_proj	16,777,216	3,445,710,848
model.layers.10.self_attn.v_proj	4,194,304	3,449,905,152
model.layers.11.input_layernorm	4,096	3,449,909,248
model.layers.11.mlp.down_proj	58,720,256	3,508,629,504
model.layers.11.mlp.gate_proj	58,720,256	3,567,349,760

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Module	Params	Running total
model.layers.11.mlp.up_proj	58,720,256	3,626,070,016
model.layers.11.post_attention_layernorm	4,096	3,626,074,112
model.layers.11.self_attn.k_proj	4,194,304	3,630,268,416
model.layers.11.self_attn.o_proj	16,777,216	3,647,045,632
model.layers.11.self_attn.q_proj	16,777,216	3,663,822,848
model.layers.11.self_attn.v_proj	4,194,304	3,668,017,152
model.layers.12.input_layernorm	4,096	3,668,021,248
model.layers.12.mlp.down_proj	58,720,256	3,726,741,504
model.layers.12.mlp.gate_proj	58,720,256	3,785,461,760
model.layers.12.mlp.up_proj	58,720,256	3,844,182,016
model.layers.12.post_attention_layernorm	4,096	3,844,186,112
model.layers.12.self_attn.k_proj	4,194,304	3,848,380,416
model.layers.12.self_attn.o_proj	16,777,216	3,865,157,632
model.layers.12.self_attn.q_proj	16,777,216	3,881,934,848
model.layers.12.self_attn.v_proj	4,194,304	3,886,129,152
model.layers.13.input_layernorm	4,096	3,886,133,248
model.layers.13.mlp.down_proj	58,720,256	3,944,853,504
model.layers.13.mlp.gate_proj	58,720,256	4,003,573,760
model.layers.13.mlp.up_proj	58,720,256	4,062,294,016
model.layers.13.post_attention_layernorm	4,096	4,062,298,112
model.layers.13.self_attn.k_proj	4,194,304	4,066,492,416
model.layers.13.self_attn.o_proj	16,777,216	4,083,269,632
model.layers.13.self_attn.q_proj	16,777,216	4,100,046,848
model.layers.13.self_attn.v_proj	4,194,304	4,104,241,152
model.layers.14.input_layernorm	4,096	4,104,245,248
model.layers.14.mlp.down_proj	58,720,256	4,162,965,504
model.layers.14.mlp.gate_proj	58,720,256	4,221,685,760
model.layers.14.mlp.up_proj	58,720,256	4,280,406,016
model.layers.14.post_attention_layernorm	4,096	4,280,410,112
model.layers.14.self_attn.k_proj	4,194,304	4,284,604,416
model.layers.14.self_attn.o_proj	16,777,216	4,301,381,632
model.layers.14.self_attn.q_proj	16,777,216	4,318,158,848
model.layers.14.self_attn.v_proj	4,194,304	4,322,353,152
model.layers.15.input_layernorm	4,096	4,322,357,248
model.layers.15.mlp.down_proj	58,720,256	4,381,077,504
model.layers.15.mlp.gate_proj	58,720,256	4,439,797,760
model.layers.15.mlp.up_proj	58,720,256	4,498,518,016
model.layers.15.post_attention_layernorm	4,096	4,498,522,112
model.layers.15.self_attn.k_proj	4,194,304	4,502,716,416
model.layers.15.self_attn.o_proj	16,777,216	4,519,493,632
model.layers.15.self_attn.q_proj	16,777,216	4,536,270,848
model.layers.15.self_attn.v_proj	4,194,304	4,540,465,152
model.layers.16.input_layernorm	4,096	4,540,469,248
model.layers.16.mlp.down_proj	58,720,256	4,599,189,504
model.layers.16.mlp.gate_proj	58,720,256	4,657,909,760
model.layers.16.mlp.up_proj	58,720,256	4,716,630,016
model.layers.16.post_attention_layernorm	4,096	4,716,634,112
model.layers.16.self_attn.k_proj	4,194,304	4,720,828,416
model.layers.16.self_attn.o_proj	16,777,216	4,737,605,632
model.layers.16.self_attn.q_proj	16,777,216	4,754,382,848
model.layers.16.self_attn.v_proj	4,194,304	4,758,577,152
model.layers.17.input_layernorm	4,096	4,758,581,248
model.layers.17.mlp.down_proj	58,720,256	4,817,301,504
model.layers.17.mlp.gate_proj	58,720,256	4,876,021,760
model.layers.17.mlp.up_proj	58,720,256	4,934,742,016
model.layers.17.post_attention_layernorm	4,096	4,934,746,112
model.layers.17.self_attn.k_proj	4,194,304	4,938,940,416
model.layers.17.self_attn.o_proj	16,777,216	4,955,717,632

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Module	Params	Running total
model.layers.17.self_attn.q_proj	16,777,216	4,972,494,848
model.layers.17.self_attn.v_proj	4,194,304	4,976,689,152
model.layers.18.input_layernorm	4,096	4,976,693,248
model.layers.18.mlp.down_proj	58,720,256	5,035,413,504
model.layers.18.mlp.gate_proj	58,720,256	5,094,133,760
model.layers.18.mlp.up_proj	58,720,256	5,152,854,016
model.layers.18.post_attention_layernorm	4,096	5,152,858,112
model.layers.18.self_attn.k_proj	4,194,304	5,157,052,416
model.layers.18.self_attn.o_proj	16,777,216	5,173,829,632
model.layers.18.self_attn.q_proj	16,777,216	5,190,606,848
model.layers.18.self_attn.v_proj	4,194,304	5,194,801,152
model.layers.19.input_layernorm	4,096	5,194,805,248
model.layers.19.mlp.down_proj	58,720,256	5,253,525,504
model.layers.19.mlp.gate_proj	58,720,256	5,312,245,760
model.layers.19.mlp.up_proj	58,720,256	5,370,966,016
model.layers.19.post_attention_layernorm	4,096	5,370,970,112
model.layers.19.self_attn.k_proj	4,194,304	5,375,164,416
model.layers.19.self_attn.o_proj	16,777,216	5,391,941,632
model.layers.19.self_attn.q_proj	16,777,216	5,408,718,848
model.layers.19.self_attn.v_proj	4,194,304	5,412,913,152
model.layers.20.input_layernorm	4,096	5,412,917,248
model.layers.20.mlp.down_proj	58,720,256	5,471,637,504
model.layers.20.mlp.gate_proj	58,720,256	5,530,357,760
model.layers.20.mlp.up_proj	58,720,256	5,589,078,016
model.layers.20.post_attention_layernorm	4,096	5,589,082,112
model.layers.20.self_attn.k_proj	4,194,304	5,593,276,416
model.layers.20.self_attn.o_proj	16,777,216	5,610,053,632
model.layers.20.self_attn.q_proj	16,777,216	5,626,830,848
model.layers.20.self_attn.v_proj	4,194,304	5,631,025,152
model.layers.21.input_layernorm	4,096	5,631,029,248
model.layers.21.mlp.down_proj	58,720,256	5,689,749,504
model.layers.21.mlp.gate_proj	58,720,256	5,748,469,760
model.layers.21.mlp.up_proj	58,720,256	5,807,190,016
model.layers.21.post_attention_layernorm	4,096	5,807,194,112
model.layers.21.self_attn.k_proj	4,194,304	5,811,388,416
model.layers.21.self_attn.o_proj	16,777,216	5,828,165,632
model.layers.21.self_attn.q_proj	16,777,216	5,844,942,848
model.layers.21.self_attn.v_proj	4,194,304	5,849,137,152
model.layers.22.input_layernorm	4,096	5,849,141,248
model.layers.22.mlp.down_proj	58,720,256	5,907,861,504
model.layers.22.mlp.gate_proj	58,720,256	5,966,581,760
model.layers.22.mlp.up_proj	58,720,256	6,025,302,016
model.layers.22.post_attention_layernorm	4,096	6,025,306,112
model.layers.22.self_attn.k_proj	4,194,304	6,029,500,416
model.layers.22.self_attn.o_proj	16,777,216	6,046,277,632
model.layers.22.self_attn.q_proj	16,777,216	6,063,054,848
model.layers.22.self_attn.v_proj	4,194,304	6,067,249,152
model.layers.23.input_layernorm	4,096	6,067,253,248
model.layers.23.mlp.down_proj	58,720,256	6,125,973,504
model.layers.23.mlp.gate_proj	58,720,256	6,184,693,760
model.layers.23.mlp.up_proj	58,720,256	6,243,414,016
model.layers.23.post_attention_layernorm	4,096	6,243,418,112
model.layers.23.self_attn.k_proj	4,194,304	6,247,612,416
model.layers.23.self_attn.o_proj	16,777,216	6,264,389,632
model.layers.23.self_attn.q_proj	16,777,216	6,281,166,848
model.layers.23.self_attn.v_proj	4,194,304	6,285,361,152
model.layers.24.input_layernorm	4,096	6,285,365,248
model.layers.24.mlp.down_proj	58,720,256	6,344,085,504

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Module	Params	Running total
model.layers.24.mlp.gate_proj	58,720,256	6,402,805,760
model.layers.24.mlp.up_proj	58,720,256	6,461,526,016
model.layers.24.post_attention_layernorm	4,096	6,461,530,112
model.layers.24.self_attn.k_proj	4,194,304	6,465,724,416
model.layers.24.self_attn.o_proj	16,777,216	6,482,501,632
model.layers.24.self_attn.q_proj	16,777,216	6,499,278,848
model.layers.24.self_attn.v_proj	4,194,304	6,503,473,152
model.layers.25.input_layernorm	4,096	6,503,477,248
model.layers.25.mlp.down_proj	58,720,256	6,562,197,504
model.layers.25.mlp.gate_proj	58,720,256	6,620,917,760
model.layers.25.mlp.up_proj	58,720,256	6,679,638,016
model.layers.25.post_attention_layernorm	4,096	6,679,642,112
model.layers.25.self_attn.k_proj	4,194,304	6,683,836,416
model.layers.25.self_attn.o_proj	16,777,216	6,700,613,632
model.layers.25.self_attn.q_proj	16,777,216	6,717,390,848
model.layers.25.self_attn.v_proj	4,194,304	6,721,585,152
model.layers.26.input_layernorm	4,096	6,721,589,248
model.layers.26.mlp.down_proj	58,720,256	6,780,309,504
model.layers.26.mlp.gate_proj	58,720,256	6,839,029,760
model.layers.26.mlp.up_proj	58,720,256	6,897,750,016
model.layers.26.post_attention_layernorm	4,096	6,897,754,112
model.layers.26.self_attn.k_proj	4,194,304	6,901,948,416
model.layers.26.self_attn.o_proj	16,777,216	6,918,725,632
model.layers.26.self_attn.q_proj	16,777,216	6,935,502,848
model.layers.26.self_attn.v_proj	4,194,304	6,939,697,152
model.layers.27.input_layernorm	4,096	6,939,701,248
model.layers.27.mlp.down_proj	58,720,256	6,998,421,504
model.layers.27.mlp.gate_proj	58,720,256	7,057,141,760
model.layers.27.mlp.up_proj	58,720,256	7,115,862,016
model.layers.27.post_attention_layernorm	4,096	7,115,866,112
model.layers.27.self_attn.k_proj	4,194,304	7,120,060,416
model.layers.27.self_attn.o_proj	16,777,216	7,136,837,632
model.layers.27.self_attn.q_proj	16,777,216	7,153,614,848
model.layers.27.self_attn.v_proj	4,194,304	7,157,809,152
model.layers.28.input_layernorm	4,096	7,157,813,248
model.layers.28.mlp.down_proj	58,720,256	7,216,533,504
model.layers.28.mlp.gate_proj	58,720,256	7,275,253,760
model.layers.28.mlp.up_proj	58,720,256	7,333,974,016
model.layers.28.post_attention_layernorm	4,096	7,333,978,112
model.layers.28.self_attn.k_proj	4,194,304	7,338,172,416
model.layers.28.self_attn.o_proj	16,777,216	7,354,949,632
model.layers.28.self_attn.q_proj	16,777,216	7,371,726,848
model.layers.28.self_attn.v_proj	4,194,304	7,375,921,152
model.layers.29.input_layernorm	4,096	7,375,925,248
model.layers.29.mlp.down_proj	58,720,256	7,434,645,504
model.layers.29.mlp.gate_proj	58,720,256	7,493,365,760
model.layers.29.mlp.up_proj	58,720,256	7,552,086,016
model.layers.29.post_attention_layernorm	4,096	7,552,090,112
model.layers.29.self_attn.k_proj	4,194,304	7,556,284,416
model.layers.29.self_attn.o_proj	16,777,216	7,573,061,632
model.layers.29.self_attn.q_proj	16,777,216	7,589,838,848
model.layers.29.self_attn.v_proj	4,194,304	7,594,033,152
model.layers.30.input_layernorm	4,096	7,594,037,248
model.layers.30.mlp.down_proj	58,720,256	7,652,757,504
model.layers.30.mlp.gate_proj	58,720,256	7,711,477,760
model.layers.30.mlp.up_proj	58,720,256	7,770,198,016
model.layers.30.post_attention_layernorm	4,096	7,770,202,112
model.layers.30.self_attn.k_proj	4,194,304	7,774,396,416

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Module	Params	Running total
model.layers.30.self_attn.o_proj	16,777,216	7,791,173,632
model.layers.30.self_attn.q_proj	16,777,216	7,807,950,848
model.layers.30.self_attn.v_proj	4,194,304	7,812,145,152
model.layers.31.input_layernorm	4,096	7,812,149,248
model.layers.31.mlp.down_proj	58,720,256	7,870,869,504
model.layers.31.mlp.gate_proj	58,720,256	7,929,589,760
model.layers.31.mlp.up_proj	58,720,256	7,988,310,016
model.layers.31.post_attention_layernorm	4,096	7,988,314,112
model.layers.31.self_attn.k_proj	4,194,304	7,992,508,416
model.layers.31.self_attn.o_proj	16,777,216	8,009,285,632
model.layers.31.self_attn.q_proj	16,777,216	8,026,062,848
model.layers.31.self_attn.v_proj	4,194,304	8,030,257,152
model.norm	4,096	8,030,261,248